

Topics for week 4

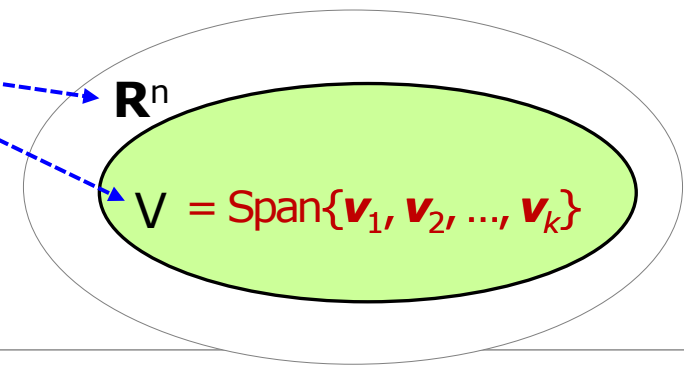
Chapter 3: Linear Transformation, Eigenvalues and Eigenvectors

- 3.1 Linear Transformation
- 3.2 Eigenvalues and Eigenvectors
- 3.3 Eigenspaces
- 3.4 Diagonalizable Matrices

A collection of objects that you can perform addition and scalar multiplication and satisfy the closure properties.

May not have the same dimension

$$\dim V \leq \dim \mathbf{R}^n = n$$



Vector space (chapter 2)

- In this course, vector space refers to $\mathbf{R}^n$ and any subspace V of $\mathbf{R}^n$.
- They can all be written as $\text{Span}\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k\}$ where $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k$ belongs to $\mathbf{R}^n$.
- Subspaces we learn in this course: Say $\mathbf{A}$ is an $n \times m$ matrix

▪ Solution space of $\mathbf{Ax} = \mathbf{0}$

subspace of $\mathbf{R}^m$

▪ Row space of a matrix $\mathbf{A}$

subspace of $\mathbf{R}^m$

▪ Column space of a matrix $\mathbf{A}$

subspace of $\mathbf{R}^n$

▪ Nullspace of a matrix $\mathbf{A}$

subspace of $\mathbf{R}^m$

▪ Eigenspace associated with eigenvalue λ of a matrix $\mathbf{A}$

▪ A line in $\mathbf{R}^2$ or $\mathbf{R}^3$ that passes through the origin

subspace of $\mathbf{R}^2$ or $\mathbf{R}^3$

▪ A plane in $\mathbf{R}^3$ that contains the origin

subspace of $\mathbf{R}^3$

Matrix as Mapping

$$f : \mathbf{R} \rightarrow \mathbf{R}$$

$$f(x) = \sqrt{x}$$

$$\mathbf{A} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \quad \mathbf{u} = \begin{pmatrix} x \\ y \end{pmatrix} \quad \mathbf{A}\mathbf{u} = \begin{pmatrix} -y \\ x \end{pmatrix}$$

input vector

output vector

$$\mathbf{u} \longrightarrow T(\mathbf{u}) \quad \text{image of } \mathbf{u} \text{ under } T$$

$\mathbf{A}$ can be regarded as a mapping or transformation

$T : \mathbf{R}^2 \rightarrow \mathbf{R}^2$ (a mapping or function)

$$T\left(\begin{pmatrix} x \\ y \end{pmatrix}\right) = \begin{pmatrix} -y \\ x \end{pmatrix}$$

$$\text{for all } \begin{pmatrix} x \\ y \end{pmatrix} \in \mathbf{R}^2$$

T : linear transformation from $\mathbf{R}^2$ to $\mathbf{R}^2$

Linear Transformation

A: $m \times n$ matrix $\rightarrow$ **standard matrix** of the linear transformation

$T : \mathbf{R}^n \rightarrow \mathbf{R}^m$ defined by $T(\mathbf{u}) = \mathbf{A}\mathbf{u}$ for all $\mathbf{u} \in \mathbf{R}^n$

T is called a **linear transformation** from $\mathbf{R}^n$ to $\mathbf{R}^m$

Linearity Conditions

1. $T(\mathbf{0}) = \mathbf{0}$
2. $T(\mathbf{u} + \mathbf{v}) = T(\mathbf{u}) + T(\mathbf{v})$
3. $T(c\mathbf{u}) = cT(\mathbf{u})$
4. $T(c_1\mathbf{u}_1 + c_2\mathbf{u}_2 + \cdots + c_k\mathbf{u}_k) = c_1T(\mathbf{u}_1) + c_2T(\mathbf{u}_2) + \cdots + c_kT(\mathbf{u}_k)$

Example 1:

Standard matrix of linear transformation

Find the standard matrices of the following linear transformations

$$T : \mathbf{R}^3 \rightarrow \mathbf{R}^2$$

$$T \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} x + 2y \\ 3x - y + 5z \end{pmatrix} = x \begin{pmatrix} 1 \\ 3 \end{pmatrix} + y \begin{pmatrix} 2 \\ -1 \end{pmatrix} + z \begin{pmatrix} 0 \\ 5 \end{pmatrix} = \boxed{\begin{pmatrix} 1 & 2 & 0 \\ 3 & -1 & 5 \end{pmatrix}} \begin{pmatrix} x \\ y \\ z \end{pmatrix}$$

Formula of T Linear combination Matrix multiplication standard matrix of T

$$S : \mathbf{R}^2 \rightarrow \mathbf{R}^3$$

$$S \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x + y \\ 0 \\ y - 2x \end{pmatrix} = \boxed{\begin{pmatrix} 1 & 1 \\ 0 & 0 \\ -2 & 1 \end{pmatrix}} \begin{pmatrix} x \\ y \end{pmatrix}$$

Formula of S standard matrix of S

Matrix multiplication by columns

Rules of matrix multiplication

$$AB = C \rightarrow B = A^{-1}C$$

$$AB = C \rightarrow A = CB^{-1}$$

$$A \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix} \quad A \begin{pmatrix} y_1 \\ y_2 \\ y_3 \end{pmatrix} = \begin{pmatrix} b_1 \\ b_2 \\ b_3 \end{pmatrix} \quad A \begin{pmatrix} z_1 \\ z_2 \\ z_3 \end{pmatrix} = \begin{pmatrix} c_1 \\ c_2 \\ c_3 \end{pmatrix}$$

Works for
any sizes

stacking

splitting

$$A \begin{pmatrix} x_1 & y_1 & z_1 \\ x_2 & y_2 & z_2 \\ x_3 & y_3 & z_3 \end{pmatrix} = \begin{pmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{pmatrix}$$

$$A = \begin{pmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{pmatrix} \begin{pmatrix} x_1 & y_1 & z_1 \\ x_2 & y_2 & z_2 \\ x_3 & y_3 & z_3 \end{pmatrix}^{-1}$$

← provided this matrix is non-singular

Given

$$A \begin{pmatrix} 1 \\ 2 \end{pmatrix} = \begin{pmatrix} 5 \\ 7 \end{pmatrix}, \quad A \begin{pmatrix} 2 \\ 3 \end{pmatrix} = \begin{pmatrix} 6 \\ 8 \end{pmatrix}$$

Find **A**.

$$A \begin{pmatrix} 1 & 2 \\ 2 & 3 \end{pmatrix} = \begin{pmatrix} 5 & 6 \\ 7 & 8 \end{pmatrix}$$

need to
post-multiply
the inverse

$$A = \begin{pmatrix} 5 & 6 \\ 7 & 8 \end{pmatrix} \begin{pmatrix} 1 & 2 \\ 2 & 3 \end{pmatrix}^{-1} = \begin{pmatrix} -3 & 4 \\ -5 & 6 \end{pmatrix}$$

Example 2: Image of basis under linear transformation

$T : \mathbf{R}^3 \rightarrow \mathbf{R}^3$ is a **linear transformation**

Given $\mathbf{A} \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix} = \begin{pmatrix} 2 \\ 6 \\ 6 \end{pmatrix}$, $\mathbf{A} \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 4 \\ 8 \\ 2 \end{pmatrix}$ and $\mathbf{A} \begin{pmatrix} 0 \\ 2 \\ 0 \end{pmatrix} = \begin{pmatrix} 6 \\ 6 \\ 6 \end{pmatrix}$. Find the standard matrix $\mathbf{A}$ for T .

$$\mathbf{A} \begin{pmatrix} 1 & 1 & 0 \\ 2 & 1 & 2 \\ 1 & 0 & 0 \end{pmatrix} = \begin{pmatrix} 2 & 4 & 6 \\ 6 & 8 & 6 \\ 6 & 2 & 6 \end{pmatrix}$$



$$\mathbf{A} = \begin{pmatrix} 2 & 4 & 6 \\ 6 & 8 & 6 \\ 6 & 2 & 6 \end{pmatrix} \begin{pmatrix} 1 & 1 & 0 \\ 2 & 1 & 2 \\ 1 & 0 & 0 \end{pmatrix}^{-1}$$

$$\mathbf{A} = \begin{pmatrix} 1 & 3 & -5 \\ 5 & 3 & -5 \\ -1 & 3 & 1 \end{pmatrix}$$

$$\begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & -1 \\ -0.5 & 0.5 & -0.5 \end{pmatrix}$$

check non-singular

the 3 columns are linearly independent



$\begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 2 \\ 0 \end{pmatrix}$ form a basis for $\mathbf{R}^3$

Linear Transformation

$\{\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_n\}$: a basis for $\mathbf{R}^n$

$T : \mathbf{R}^n \rightarrow \mathbf{R}^m$ is a linear transformation

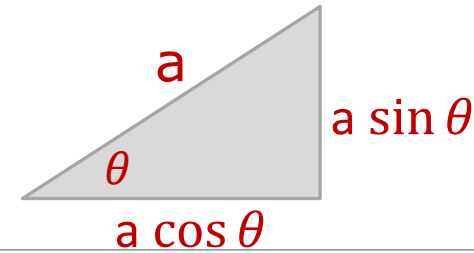
If we know the images (output vectors) $T(\mathbf{u}_1), T(\mathbf{u}_2), \dots, T(\mathbf{u}_n)$,
we have enough information to

- find the standard matrix of T
- determine the image $T(\mathbf{v})$ of any vector $\mathbf{v}$ in the domain $\mathbf{R}^n$.

$\{\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_n\}$: standard basis for $\mathbf{R}^n$

$$\mathbf{A} = \left(T(\mathbf{e}_1) \ T(\mathbf{e}_2) \ \dots \ T(\mathbf{e}_n) \right)$$

Example 3: Image of basis under linear transformation



Revise your trigonometry

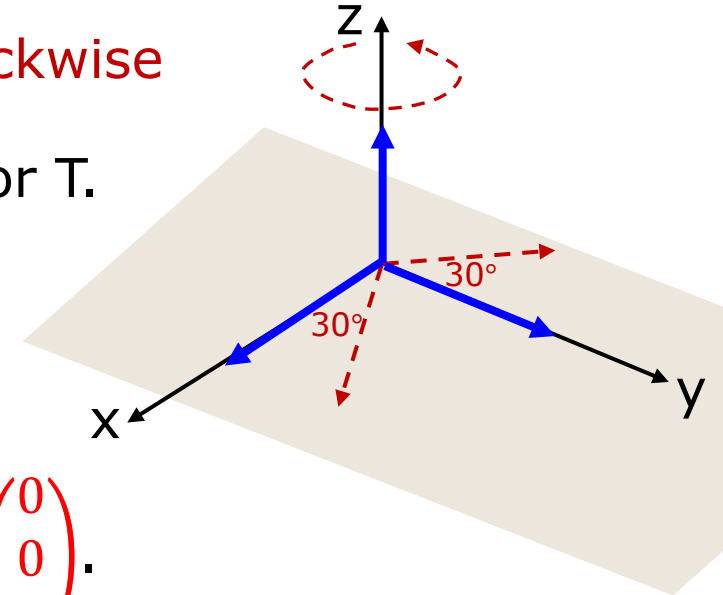
$T : \mathbf{R}^3 \rightarrow \mathbf{R}^3$ is a rotation about the z-axis with 30° anticlockwise

T is a linear transformation Find the standard matrix for T .

Use the standard basis $\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$ for $\mathbf{R}^3$

We have $\mathbf{A} \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} \frac{\sqrt{3}}{2} \\ \frac{1}{2} \\ 0 \end{pmatrix}$, $\mathbf{A} \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} = \begin{pmatrix} -\frac{1}{2} \\ \frac{\sqrt{3}}{2} \\ 0 \end{pmatrix}$ and $\mathbf{A} \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$.

$$\mathbf{A} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} \frac{\sqrt{3}}{2} & -\frac{1}{2} & 0 \\ \frac{1}{2} & \frac{\sqrt{3}}{2} & 0 \\ 0 & 0 & 1 \end{pmatrix} \longrightarrow \mathbf{A} = \begin{pmatrix} \frac{\sqrt{3}}{2} & -\frac{1}{2} & 0 \\ \frac{1}{2} & \frac{\sqrt{3}}{2} & 0 \\ 0 & 0 & 1 \end{pmatrix} \text{ is the standard matrix of } T$$



Example 4: Image of basis under linear transformation

$T : \mathbf{R}^3 \rightarrow \mathbf{R}^3$ is a **linear transformation**

Suppose we know the images of $T\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$ and $T\begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$ but not $T\begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$.

Yes

No

Can we find the images of $T\begin{pmatrix} 2 \\ 3 \\ 0 \end{pmatrix}$ and $T\begin{pmatrix} 0 \\ 2 \\ 3 \end{pmatrix}$?

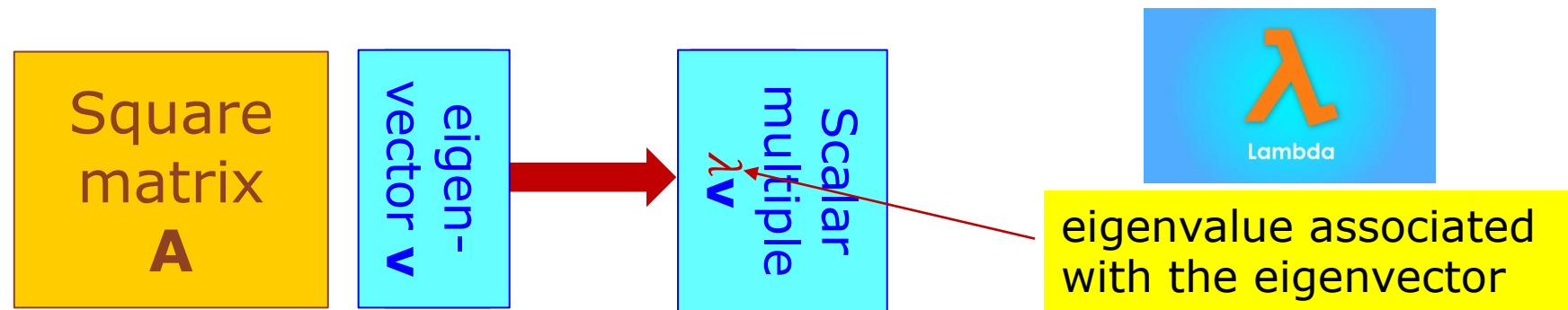
Linearity Condition

$$T(c_1 \mathbf{u}_1 + c_2 \mathbf{u}_2 + \cdots + c_k \mathbf{u}_k) = c_1 T(\mathbf{u}_1) + c_2 T(\mathbf{u}_2) + \cdots + c_k T(\mathbf{u}_k)$$

$$\begin{pmatrix} 2 \\ 3 \\ 0 \end{pmatrix} = 2 \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} + 3 \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} \Rightarrow T\begin{pmatrix} 2 \\ 3 \\ 0 \end{pmatrix} = T\left[2 \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} + 3 \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}\right] = 2T\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} + 3T\begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$$

$\begin{pmatrix} 0 \\ 2 \\ 3 \end{pmatrix}$ is not a linear combination of $\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$ and $\begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$

Eigenvalue and Eigenvector visualization



When input and output vectors
are scalar multiples

Example 5: Identifying eigenvectors

$$\mathbf{A} = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 1 \end{pmatrix}$$

Are these eigenvectors of $\mathbf{A}$: $\begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}$?

$$\begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix}$$

$\begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix}$ is an eigenvector with eigenvalue 1

$$\begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 2 \\ 0 \\ 2 \end{pmatrix}$$

$\begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}$ is an eigenvector with eigenvalue 2

$$\begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix}$$

$\begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}$ is not an eigenvector

$$\begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

$\begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}$ is an eigenvector with eigenvalue 0

Ways To Find Eigenvalues

- If an eigenvector $\mathbf{u}$ is given, multiply it by the matrix: $\mathbf{A}\mathbf{u} = \lambda\mathbf{u}$
- Solve characteristic equation $\det(\lambda\mathbf{I} - \mathbf{A}) = 0$
- If the matrix is triangular, take the diagonal entries
- If λ is an eigenvalue of $\mathbf{A}$, then
 - λ is an eigenvalue of $\mathbf{A}^T$
 - λ^n is an eigenvalue of $\mathbf{A}^n$
 - λ^{-1} is an eigenvalue of $\mathbf{A}^{-1}$ (when $\mathbf{A}$ is non-singular)

Example 6: Finding eigenvalues from characteristic polynomials

$$\mathbf{A} = \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix}$$

Characteristic polynomial:

$$\det(\lambda \mathbf{I} - \mathbf{A}) = \begin{vmatrix} \lambda & 0 & -1 & 0 \\ 0 & \lambda & 0 & -1 \\ -1 & 0 & \lambda & 0 \\ 0 & -1 & 0 & \lambda \end{vmatrix}$$

$$= \lambda \times \begin{vmatrix} \lambda & 0 & -1 \\ 0 & \lambda & 0 \\ -1 & 0 & \lambda \end{vmatrix} - 1 \times \begin{vmatrix} 0 & \lambda & -1 \\ -1 & 0 & 0 \\ 0 & -1 & \lambda \end{vmatrix}$$

$$= \lambda(\lambda^3 - \lambda) - (1)(\lambda^2 - 1)$$

$$= \lambda^2(\lambda^2 - 1) - (1)(\lambda^2 - 1)$$

$$= (\lambda^2 - 1)(\lambda^2 - 1)$$

$$= (\lambda - 1)(\lambda + 1)(\lambda - 1)(\lambda + 1) = (\lambda - 1)^2(\lambda + 1)^2$$

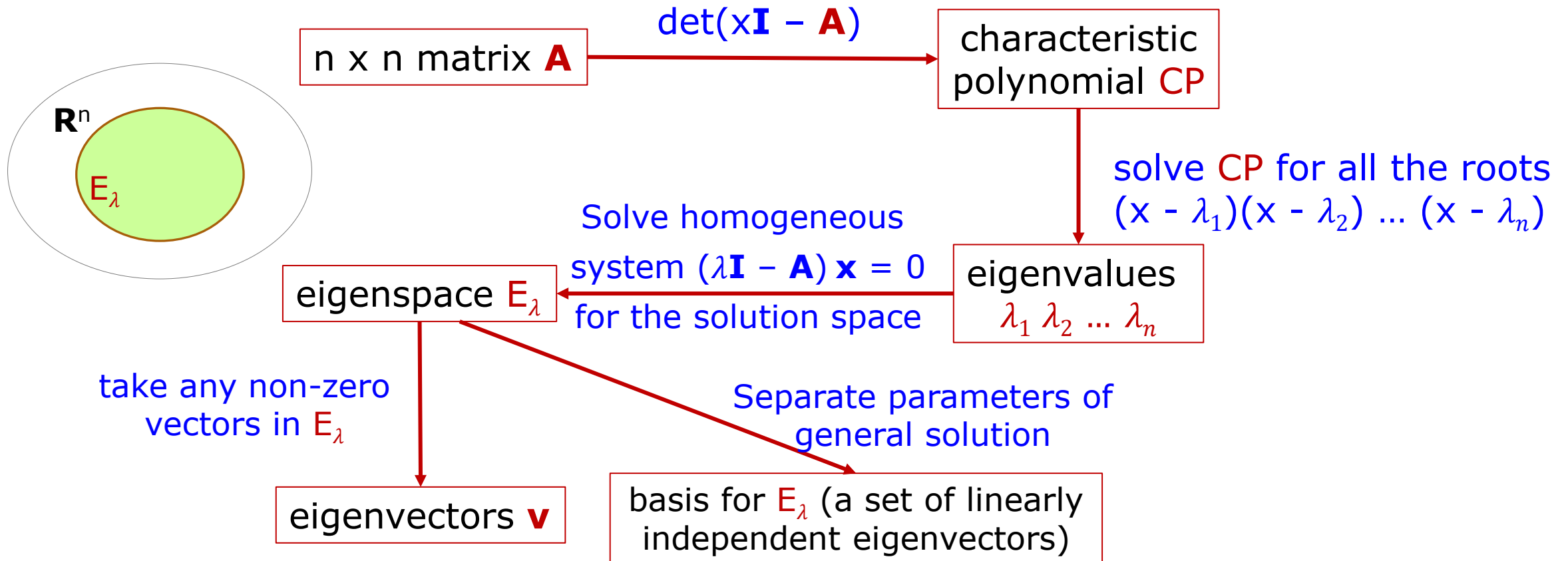
Multiplicities of the eigenvalues

So the eigenvalues of $\mathbf{A}$ are 1 and -1.



The roots of the polynomial are 1 and -1.

Finding Eigenvalue and Eigenvector



Check: Eigenspace for $\lambda = -1$: $E_{-1} = \text{span}\left\{\begin{pmatrix} -1 \\ 0 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ -1 \\ 0 \\ 1 \end{pmatrix}\right\}$

$\dim E_{-1} = 2$

Finding eigenvector (given eigenvalue)

$$A = \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix}$$

Solve $(\lambda I - A) x = 0$ for eigenvalues $\lambda = 1$ and -1

$\lambda = 1$

$$\begin{pmatrix} \lambda & 0 & -1 & 0 \\ 0 & \lambda & 0 & -1 \\ -1 & 0 & \lambda & 0 \\ 0 & -1 & 0 & \lambda \end{pmatrix} \begin{pmatrix} x \\ y \\ z \\ w \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \end{pmatrix}$$

$$\left(\begin{array}{cccc|c} 1 & 0 & -1 & 0 & 0 \\ 0 & 1 & 0 & -1 & 0 \\ -1 & 0 & 1 & 0 & 0 \\ 0 & -1 & 0 & 1 & 0 \end{array} \right) \xrightarrow{G.E.} \left(\begin{array}{cccc|c} 1 & 0 & -1 & 0 & 0 \\ 0 & 1 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{array} \right)$$

$w = t, z = s, y = t, x = s$

$\dim E_1 = 2$

Basis for Eigenspace E_1

Gen. soln: $\begin{pmatrix} x \\ y \\ z \\ w \end{pmatrix} = \begin{pmatrix} s \\ t \\ s \\ t \end{pmatrix} = s \begin{pmatrix} 1 \\ 0 \\ 1 \\ 0 \end{pmatrix} + t \begin{pmatrix} 0 \\ 1 \\ 0 \\ 1 \end{pmatrix}$

Eigenspace for $\lambda = 1$: $E_1 = \text{span}\left\{\begin{pmatrix} 1 \\ 0 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 0 \\ 1 \end{pmatrix}\right\}$

Any linear combination is an eigenvector associated to $\lambda = 1$

The dimension of eigenspace can be equal but cannot be more than the multiplicity of the eigenvalue in the characteristic polynomial

Multiplicity and dimension of eigenspace

$$\mathbf{A} = \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix}$$

Characteristic polynomial $(\lambda - 1)^2(\lambda + 1)^2$

Eigenvalues of $\mathbf{A}$ are 1 and -1

Eigenspaces of $\mathbf{A}$ are E_1 and E_{-1} :

$$E_1 = \text{span} \left\{ \begin{pmatrix} 1 \\ 0 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 0 \\ 1 \end{pmatrix} \right\}$$

$$\dim E_1 = 2$$

$$E_{-1} = \text{span} \left\{ \begin{pmatrix} -1 \\ 0 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ -1 \\ 0 \\ 1 \end{pmatrix} \right\}$$

$$\dim E_{-1} = 2$$

Example

Characteristic polynomial of $\mathbf{B}$: $(\lambda - 1)^1(\lambda - 2)^3(\lambda - 4)^2$

Eigenvalues of $\mathbf{B}$ are 1, 2 and 4 with multiplicity 1, 3, 2 respectively

Eigenspaces of $\mathbf{B}$ are E_1 , E_2 and E_4 :

$$\dim E_1 = 1 \quad \dim E_2 \leq 3 \quad \dim E_4 \leq 2$$

Lecture Quiz 2

$$\mathbf{A} = \begin{pmatrix} 4 & a & b & c & d \\ 0 & 4 & e & f & g \\ 0 & 0 & 4 & h & i \\ 0 & 0 & 0 & 5 & j \\ 0 & 0 & 0 & 0 & 5 \end{pmatrix} \quad 4 \text{ and } 5 \text{ are the \textcolor{red}{only} eigenvalues of } \mathbf{A}.$$

What are the possible dimensions of the eigenspaces E_4 and E_5 ?

The characteristic polynomial of $\mathbf{A}$: $(x - 4)^3(x - 5)^2$

$\dim E_4 = 1, 2 \text{ or } 3$ and $\dim E_5 = 1, 2$

$$\mathbf{P}^{-1} \mathbf{A} \mathbf{P} \longrightarrow \mathbf{D}$$

diagonalizable matrix diagonal matrix

Diagonalizable Matrices

An $n \times n$ square matrix $\mathbf{A}$ is called **diagonalizable** if we can find a non-singular matrix $\mathbf{P}$ such that $\mathbf{P}^{-1}\mathbf{A}\mathbf{P}$ is a diagonal matrix.

Example of **diagonalizable** matrices

$$\begin{pmatrix} 5 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 3 \end{pmatrix}$$

All **diagonal** matrices are diagonalizable

$$\begin{pmatrix} 1 & 1 & 1 \\ 0 & 2 & 1 \\ 0 & 0 & 3 \end{pmatrix}$$

All $n \times n$ matrices with **n distinct eigenvalues** are diagonalizable

$$\begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix}$$

All $n \times n$ matrices with **n linearly independent eigenvectors** are diagonalizable

Example 8:

Determine diagonalizable matrices

$$\mathbf{A} = \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix} \quad \text{eigenvalues } \lambda = 1, -1$$
$$E_1 = \text{span} \left\{ \begin{pmatrix} 1 \\ 0 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 0 \\ 1 \end{pmatrix} \right\} \quad E_{-1} = \text{span} \left\{ \begin{pmatrix} -1 \\ 0 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ -1 \\ 0 \\ 1 \end{pmatrix} \right\}$$

$\dim E_1 = 2$ $\dim E_{-1} = 2$

A is 4 x 4, and it has 4 linearly independent eigenvectors.

So **A** is diagonalizable.

$$\mathbf{B} = \begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix} \quad \text{eigenvalue } \lambda = 1$$
$$E_1 = \text{span} \left\{ \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \right\}$$

$\dim E_1 = 1$

B is 3 x 3, and it has only 1 linearly independent eigenvector.

So **B** is not diagonalizable.

Example 8:

Determine diagonalizable matrices

$$\mathbf{A} = \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix} \quad \text{eigenvalues } \lambda = 1, -1$$
$$\mathbf{E}_1 = \text{span} \left\{ \begin{pmatrix} 1 \\ 0 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 0 \\ 1 \end{pmatrix} \right\} \quad \mathbf{E}_{-1} = \text{span} \left\{ \begin{pmatrix} -1 \\ 0 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ -1 \\ 0 \\ 1 \end{pmatrix} \right\}$$

A is diagonalizable

$$\mathbf{P}^{-1} \mathbf{A} \mathbf{P} = \mathbf{D} \quad \longleftarrow \quad \mathbf{P} = \begin{pmatrix} 1 & 0 & -1 & 0 \\ 0 & 1 & 0 & -1 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \end{pmatrix} \quad \mathbf{D} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix}$$

Suppose **A** is an $n \times n$ diagonalizable matrix.

Use the n linearly independent eigenvectors to form the n columns of a matrix **P**.

This non-singular **P** will diagonalize **A**.

Writing a diagonalizable matrix as $\mathbf{PDP}^{-1}$

Suppose $\mathbf{A}$ is a diagonalizable matrix

Given eigenvalues, we can form a diagonal matrix $\mathbf{D}$

Given eigenvectors, we can form a non-singular matrix $\mathbf{P}$

Then $\mathbf{P}^{-1}\mathbf{AP} = \mathbf{D}$

So we can express $\mathbf{A}$ as $\mathbf{A} = \mathbf{PDP}^{-1}$

Example

$\mathbf{A}$ is 3 x 3 (unknown), with eigenvalues 1, 2, 4

and respective eigenvectors $\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$, $\begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}$ and $\begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix}$.

$$\text{Form } \mathbf{D} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 4 \end{pmatrix}$$

$$\text{Form } \mathbf{P} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 1 & -1 \end{pmatrix}$$

$$\mathbf{A} = \mathbf{PDP}^{-1} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 1 & -1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 4 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 1 & -1 \end{pmatrix}^{-1} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 3 & -1 \\ 0 & -1 & 3 \end{pmatrix}$$